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Class: TYIT

Subject: BD
08-2026

Date: 03-

Practical -4

AIM: MongoDB CRUD Operation:

To start MongoDB “mongosh”:

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>mongosh
Current Mongosh Log ID: 6a7016acd02e4432f77c4758
Connecting to:  mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.9.2
Using MongoDB:  8.3.7
Using Mongosh:  2.9.2

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
  The server generated these startup warnings when booting
  2026-07-29T15:59:49.717+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test>
```

Show database:

```
test> show databases;
admin    40.00 KiB
config   96.00 KiB
local    68.00 KiB
test> _
```

Create database:

```
test> use Aadil;
switched to db Aadil
Aadil> _
```

Create “student” collection:

```
Aadil> db.createCollection("student");
{ ok: 1 }
```

Show collection:

```
Aadil> show collections;
student
```

Drop collection:

```
Aadil> db.student.drop();  
true
```

Then show collection, it's empty:

```
Aadil> show collections;  
  
Aadil>
```

Again create collection "std":

```
Aadil> db.createCollection("std")  
{ ok: 1 }  
Aadil>
```

Insert values:

```
Aadil> db.std.insertOne({uid:"248IT054",name:"aadil",per:72.24,age:21,subject:["DV","BD","AI"]});  
{  
  acknowledged: true,  
  insertedId: ObjectId('6a701ce1d02e4432f77c475a')  
}
```

Insert another values:

```
Aadil> db.std.insertOne({uid:"248IT053",name:"affan",per:84,age:21,subject:["MAD","BD","AI"]});  
{  
  acknowledged: true,  
  insertedId: ObjectId('6a701d94d02e4432f77c475b')  
}  
Aadil>
```

Insert many values:

```
Aadil> db.std.insertMany([  
  {  
    uid: "248IT054",  
    name: "aadil",  
    per: 72.72,  
    age: 21,  
    subject: ["DV", "BD", "AI"]  
  },  
  {  
    uid: "248IT053",  
    name: "affan",  
    per: 84,  
    age: 22,  
    subject: ["MAD", "BD", "AI"]  
  }  
]);  
{  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId('6a701f47d02e4432f77c475c'),  
    '1': ObjectId('6a701f47d02e4432f77c475d')  
  }  
}
```

Find “std” collection:

```
Aadil> db.std.find();
[
  {
    _id: ObjectId('6a701cc6d02e4432f77c4759'),
    uid: '248IT054',
    name: 'aadil',
    per: 72.24,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ]
  },
  {
    _id: ObjectId('6a701ce1d02e4432f77c475a'),
    uid: '248IT054',
    name: 'aadil',
    per: 72.24,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ]
  },
  {
    _id: ObjectId('6a701d94d02e4432f77c475b'),
    uid: '248IT053',
    name: 'affan',
    per: 84,
    age: 21,
    subject: [ 'MAD', 'BD', 'AI' ]
  },
  {
    _id: ObjectId('6a701f47d02e4432f77c475c'),
    uid: '248IT054',
    name: 'aadil',
    per: 72.72,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ]
  },
  {
    _id: ObjectId('6a701f47d02e4432f77c475d'),
    uid: '248IT053',
    name: 'affan',
    per: 84,
    age: 22,
    subject: [ 'MAD', 'BD', 'AI' ]
  }
]
Aadil>
```

Delete “std”:

```
]
Aadil> db.std.deleteOne({name:"affan"});
{ acknowledged: true, deletedCount: 1 }
Aadil> db.std.find();
[
  {
    _id: ObjectId('6a701cc6d02e4432f77c4759'),
    uid: '248IT054',
    name: 'aadil',
    per: 72.24,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ]
  },
  {
    _id: ObjectId('6a701ce1d02e4432f77c475a'),
    uid: '248IT054',
    name: 'aadil',
    per: 72.24,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ]
  },
  {
    _id: ObjectId('6a701f47d02e4432f77c475c'),
    uid: '248IT054',
    name: 'aadil',
    per: 72.72,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ]
  },
  {
    _id: ObjectId('6a701f47d02e4432f77c475d'),
    uid: '248IT053',
    name: 'affan',
    per: 84,
    age: 22,
    subject: [ 'MAD', 'BD', 'AI' ]
  }
]
Aadil>
```

Find with the help of age:

```
Aadil> db.std.find({age:22});
[
  {
    _id: ObjectId('6a701f47d02e4432f77c475d'),
    uid: '24BIT053',
    name: 'affan',
    per: 84,
    age: 22,
    subject: [ 'MAD', 'BD', 'AI' ]
  }
]
Aadil> _
```

Update on affan:

```
Aadil> db.std.updateOne({name:"affan"},{$set:{age:21,per:65}});
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
Aadil> db.std.find({age:22});
```

```
Aadil> db.std.find({name:"affan"});
[
  {
    _id: ObjectId('6a701f47d02e4432f77c475d'),
    uid: '24BIT053',
    name: 'affan',
    per: 65,
    age: 21,
    subject: [ 'MAD', 'BD', 'AI' ]
  }
]
Aadil>
```

Realtional Operators:

\$gt – greaterthan

\$eq – equal

\$ne – notequal

\$gte – greaterthan equal

\$lt – less than

\$lte – less than equal

Find who has greater than 70 percentage:

```
Aadil> db.std.find({per:{$gt:70}});
[
  {
    _id: ObjectId('6a701cc6d02e4432f77c4759'),
    uid: '24BIT054',
    name: 'aadil',
    per: 72.24,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ]
  },
]
```

Display all the students who has failed:

```
Aadil> db.std.find({per:{$lt:40}});
```

Display all the student who's name is not aadil:

```
Aadil> db.std.find({name:{$ne:"aadil"}});
[
  {
    _id: ObjectId('6a701f47d02e4432f77c475d'),
    uid: '24BIT053',
    name: 'affan',
    per: 65,
    age: 21,
    subject: [ 'MAD', 'BD', 'AI' ]
  }
]
```

Set Membership operator:

IN - In

NIN – Not In

Who has only “MAD” subject:

```
Aadil> db.std.find({subject:"MAD"});
[
  {
    _id: ObjectId('6a701f47d02e4432f77c475d'),
    uid: '24BIT053',
    name: 'affan',
    per: 65,
    age: 21,
    subject: [ 'MAD', 'BD', 'AI' ]
  }
]
```

All the documents where subject BD and DV:

```

Aadil> db.std.find({subject:{$in:["AI","BD"]}});
[
  {
    _id: ObjectId('6a701cc6d02e4432f77c4759'),
    uid: '24BIT054',
    name: 'aadil',
    per: 72.24,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ]
  },
  {
    _id: ObjectId('6a701ce1d02e4432f77c475a'),
    uid: '24BIT054',
    name: 'aadil',
    per: 72.24,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ]
  },
  {
    _id: ObjectId('6a701f47d02e4432f77c475c'),
    uid: '24BIT054',
    name: 'aadil',
    per: 72.72,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ]
  },
  {
    _id: ObjectId('6a701f47d02e4432f77c475d'),
    uid: '24BIT053',
    name: 'affan',
    per: 65,
    age: 21,
    subject: [ 'MAD', 'BD', 'AI' ]
  }
]

```

```

Aadil> db.std.find({subject:{$nin:["BD","DV"]}});
Aadil> _

```

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Age is greater than equal to 20 add Class "TYIT":

```
Aadil> db.std.updateMany({age:{$gte:20}},{$set:{class:"TYIT"}});
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 4,
  modifiedCount: 4,
  upsertedCount: 0
}
```

Create limit of 2 records:

```
Aadil> db.std.find().limit(2);
[
  {
    _id: ObjectId('6a701cc6d02e4432f77c4759'),
    uid: '24BIT054',
    name: 'aadil',
    per: 72.24,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ],
    class: 'TYIT'
  },
  {
    _id: ObjectId('6a701ce1d02e4432f77c475a'),
    uid: '24BIT054',
    name: 'aadil',
    per: 72.24,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ],
    class: 'TYIT'
  }
]
```

Projection query to exclude the fields:

1 to include:

Find only all UID:

```
Aadil> db.std.find({}, {uid:1});
[
  { _id: ObjectId('6a701cc6d02e4432f77c4759'), uid: '24BIT054' },
  { _id: ObjectId('6a701ce1d02e4432f77c475a'), uid: '24BIT054' },
  { _id: ObjectId('6a701f47d02e4432f77c475c'), uid: '24BIT054' },
  { _id: ObjectId('6a701f47d02e4432f77c475d'), uid: '24BIT053' }
]
```

Find name with percentage only:

```

]
Aadil> db.std.find({}, {name:1, per:1});
[
  {
    _id: ObjectId('6a701cc6d02e4432f77c4759'),
    name: 'aadil',
    per: 72.24
  },
  {
    _id: ObjectId('6a701ce1d02e4432f77c475a'),
    name: 'aadil',
    per: 72.24
  },
  {
    _id: ObjectId('6a701f47d02e4432f77c475c'),
    name: 'aadil',
    per: 72.72
  },
  { _id: ObjectId('6a701f47d02e4432f77c475d'), name: 'affan', per: 65 }
]

```

Find name who has only BD subject:

```

Aadil> db.std.find({subject:"BD"}, {name:1, subject:1});
[
  {
    _id: ObjectId('6a701cc6d02e4432f77c4759'),
    name: 'aadil',
    subject: [ 'DV', 'BD', 'AI' ]
  },
]

```

0 to exclude:

Display all the columns with all the fields except field “subject”:

```
Aadil> db.std.find({}, {subject:0});
[
  {
    _id: ObjectId('6a701cc6d02e4432f77c4759'),
    uid: '24BIT054',
    name: 'aadil',
    per: 72.24,
    age: 21,
    class: 'TYIT'
  },
  {
    _id: ObjectId('6a701ce1d02e4432f77c475a'),
    uid: '24BIT054',
    name: 'aadil',
    per: 72.24,
    age: 21,
    class: 'TYIT'
  },
  {
    _id: ObjectId('6a701f47d02e4432f77c475c'),
    uid: '24BIT054',
    name: 'aadil',
    per: 72.72,
    age: 21,
    class: 'TYIT'
  },
  {
    _id: ObjectId('6a701f47d02e4432f77c475d'),
    uid: '24BIT053',
    name: 'affan',
    per: 65,
    age: 21,
    class: 'TYIT'
  }
]
```

Display in ascending order 1

```
Aadil> db.std.find().sort({per:1});
[
  {
    _id: ObjectId('6a701f47d02e4432f77c475d'),
    uid: '248IT053',
    name: 'affan',
    per: 65,
    age: 21,
    subject: [ 'MAD', 'BD', 'AI' ],
    class: 'TYIT'
  },
  {
    _id: ObjectId('6a701cc6d02e4432f77c4759'),
    uid: '248IT054',
    name: 'aadil',
    per: 72.24,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ],
    class: 'TYIT'
  },
  {
    _id: ObjectId('6a701ce1d02e4432f77c475a'),
    uid: '248IT054',
    name: 'aadil',
    per: 72.24,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ],
    class: 'TYIT'
  },
  {
    _id: ObjectId('6a701f47d02e4432f77c475c'),
    uid: '248IT054',
    name: 'aadil',
    per: 72.72,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ],
    class: 'TYIT'
  }
]
```

Display in descending order -1

```
Aadil> db.std.find().sort({per:-1});
[
  {
    _id: ObjectId('6a701f47d02e4432f77c475c'),
    uid: '248IT054',
    name: 'aadil',
    per: 72.72,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ],
    class: 'TYIT'
  },
  {
    _id: ObjectId('6a701cc6d02e4432f77c4759'),
    uid: '248IT054',
    name: 'aadil',
    per: 72.24,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ],
    class: 'TYIT'
  },
  {
    _id: ObjectId('6a701ce1d02e4432f77c475a'),
    uid: '248IT054',
    name: 'aadil',
    per: 72.24,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ],
    class: 'TYIT'
  },
  {
    _id: ObjectId('6a701f47d02e4432f77c475d'),
    uid: '248IT053',
    name: 'affan',
    per: 65,
    age: 21,
    subject: [ 'MAD', 'BD', 'AI' ],
    class: 'TYIT'
  }
]
```

Multi Sort same percentage:

Name: asec:1

```
Aadil> db.std.find().sort({per:1,name:1});
[
  {
    _id: ObjectId('6a701f47d02e4432f77c475d'),
    uid: '248IT053',
    name: 'affan',
    per: 65,
    age: 21,
    subject: [ 'MAD', 'BD', 'AI' ],
    class: 'TYIT'
  },
  {
    _id: ObjectId('6a701cc6d02e4432f77c4759'),
    uid: '248IT054',
    name: 'aadil',
    per: 72.24,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ],
    class: 'TYIT'
  },
  {
    _id: ObjectId('6a701ce1d02e4432f77c475a'),
    uid: '248IT054',
    name: 'aadil',
    per: 72.24,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ],
    class: 'TYIT'
  },
  {
    _id: ObjectId('6a701f47d02e4432f77c475c'),
    uid: '248IT054',
    name: 'aadil',
    per: 72.72,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ],
    class: 'TYIT'
  }
]
```

```
Aadil> db.std.find().sort({per:-1,name:-1});
[
  {
    _id: ObjectId('6a701f47d02e4432f77c475c'),
    uid: '248IT054',
    name: 'aadil',
    per: 72.72,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ],
    class: 'TYIT'
  },
  {
    _id: ObjectId('6a701cc6d02e4432f77c4759'),
    uid: '248IT054',
    name: 'aadil',
    per: 72.24,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ],
    class: 'TYIT'
  },
  {
    _id: ObjectId('6a701ce1d02e4432f77c475a'),
    uid: '248IT054',
    name: 'aadil',
    per: 72.24,
    age: 21,
    subject: [ 'DV', 'BD', 'AI' ],
    class: 'TYIT'
  },
  {
    _id: ObjectId('6a701f47d02e4432f77c475d'),
    uid: '248IT053',
    name: 'affan',
    per: 65,
    age: 21,
    subject: [ 'MAD', 'BD', 'AI' ],
    class: 'TYIT'
  }
]
```